GENERAL DE JESUS COLLEGE

GenPay

System Overview

A closed-loop campus e-wallet and canteen payment system that connects students, faculty, parents, canteen merchants and the school's finance office through one shared digital currency: the GenCoin.

Document	System Overview
System	GenPay / EduPay, version 3.3
Currency	GenCoin (GC), 1 GC = ₱10.00
Date	November 3, 2026

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1. Introduction

GenPay is a web-based electronic wallet system built for the General De Jesus College (GJC) campus. It replaces cash at the school canteen with a digital, school-controlled currency called the **GenCoin (GC)**. One GenCoin is always worth **₱10.00**.

Students and faculty carry a wallet and pay canteen stalls by scanning QR codes. Parents load allowances and watch over their children's spending. Canteen merchants run a point-of-sale, manage inventory and staff, and convert their earnings back to cash. The school's finance office controls how much money exists in the system, handles cash coming in and going out, leases the canteen stalls, and keeps a tamper-evident audit trail of everything that happens.

1.1 Problem Statement

- Cash handling at the canteen is slow, error-prone and hard to audit.
- Parents cannot see what their children buy or limit how much they spend.
- Stall leasing, rent collection and merchant compliance are tracked manually.
- The school has no single, reliable record of money flowing on campus.

1.2 The Solution in One Sentence

> Every peso that enters GenPay is converted into GenCoins that circulate only between school-approved wallets, and the system mathematically proves after every transaction that no GenCoin was created or lost by accident.

2. Objectives and Scope

Objectives

- Provide fast, cashless payments at canteen stalls using QR codes and wallet balances.
- Give parents visibility and control over their children's wallets.
- Give merchants a complete stall-management toolkit: POS, inventory, staff, sales reports and cash-out.
- Give the finance office full control and accountability over the money supply.
- Automate the canteen stall application, awarding and leasing process.
- Keep an append-only, hash-chained audit log of all sensitive actions.

Scope

In scope	Out of scope
Wallets for students, faculty, parents and merchants	Integration with banks or online payment gateways
Cash-in and cash-out at the school cashier	Public self-registration (accounts are created by finance)
Canteen purchases, peer transfers and allowances	Use outside the GJC campus
Stall applications, leases and rent tracking	Native mobile apps (the system is a responsive website)
Enrollment-aware wallet access per school year	Full registrar / enrollment system
Fees, fee waivers, reports, backups and audit trail	Automatic scheduled jobs (tasks run when pages are used)

3. System Users and Roles

GenPay has four main portals. Every account belongs to exactly one portal, and some portals have sub-roles with extra or fewer permissions.

Portal	Sub-roles	Who they are and what they do
Student	Student, Faculty	Students and school employees. They hold a wallet, buy food, send GenCoins to classmates and request cash-ins. Faculty use the same portal but are identified by Employee ID and are not affected by enrollment status, guardians or fee waivers.
Parent	Parent	Guardians linked to one or more students. They have their own wallet, send allowances, set spending controls and receive alerts.
Merchant	Merchant Owner, Merchant Staff	Canteen stall tenants. Owners manage the whole stall and its money; staff operate the POS and inventory on the owner's behalf.
Finance	Cashier / Admin, Super Admin	The school finance office. They manage accounts, cash movement, stalls, leases, fees and school years. Super Admins additionally control the money supply, the audit log and backups.

3.1 How Users Get Accounts

- **Students and faculty** are created by finance, one at a time or in bulk through a signed CSV import. Parents can be created and linked in the same import.
- **Merchants** apply publicly for a canteen stall. When finance awards the stall, the owner account, wallet and lease are created automatically. Finance can also provision a merchant directly.
- **Merchant staff** are created by their stall owner.
- **Parents** are created by finance, then link to their children using the student's school ID.
- New accounts receive their credentials by email and must change their password on first login.

4. System Architecture

GenPay is a server-rendered PHP web application backed by a MySQL database. All four portals share one codebase and one database. Business logic that must never go wrong, such as moving money, lives in a small set of shared core classes that every portal calls.

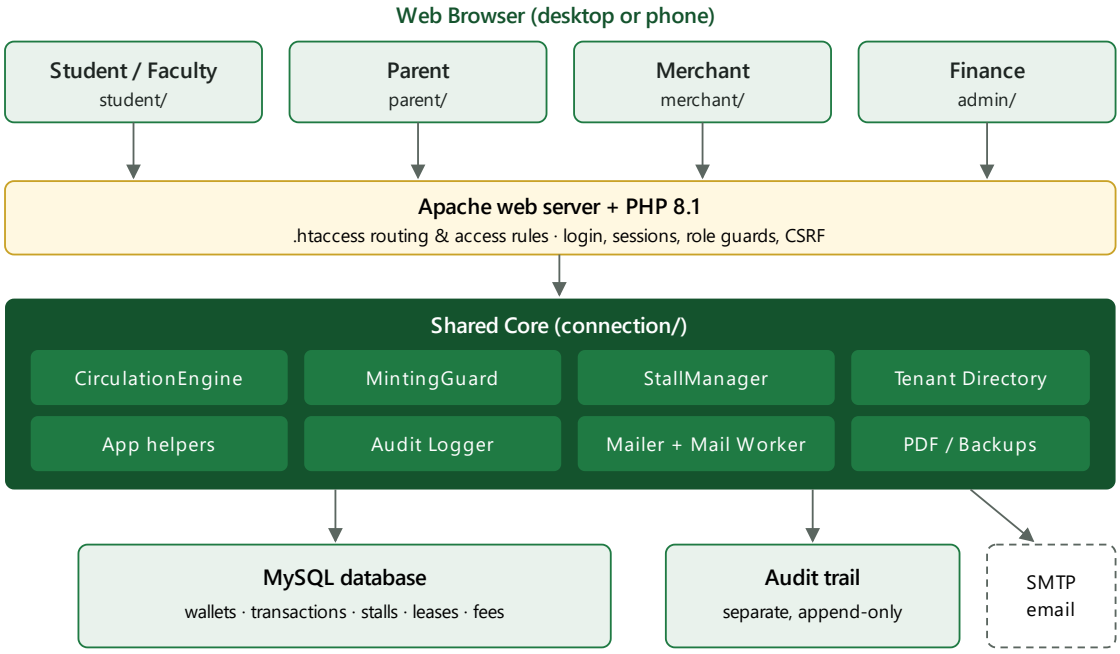


Figure 1. High-level architecture

4.1 Technology Stack

Layer	Technology
Server language	PHP 8.1 or later, no framework (plain PHP pages and JSON endpoints)
Database	MySQL / MariaDB (InnoDB, transactions, triggers) through PDO prepared statements
Web server	Apache with `.htaccess` rewrite rules (runs on Laragon / XAMPP locally)
User interface	Bootstrap 5, jQuery, DataTables, Chart.js, SweetAlert2, Font Awesome, Plus Jakarta Sans font
QR codes	QRCode.js to display codes and jsQR to scan them with the phone or laptop camera
Email	PHPMailer over SMTP, with a file-based mail queue and background worker
Documents	FPDF for printable receipts, fee forms and backups

4.2 Design Principles

- **One place for money.** Pages never change balances directly; they ask the CirculationEngine, which checks the books before saving.
- **Self-healing database.** Each feature creates or upgrades its own tables the first time it is used, so new versions deploy without manual SQL scripts.
- **Role-separated portals.** Each role has its own folder, sidebar and API endpoints, all guarded by a login and role check.
- **Everything is logged.** Sensitive actions are written to a separate, hash-chained audit trail.

5. How the System Works

5.1 The GenCoin Closed Loop

GenPay works like a small, closed economy owned by the school. A fixed amount of GenCoin is authorized to exist, called the **circulation cap**. Coins that are not in anybody's wallet sit in the **cashier vault**. Coins only move between the vault and the wallets, never in or out of thin air. After every transaction the system checks this equation:

```
Circulation Cap = Cashier Vault + Student Wallets + Merchant Wallets + Parent Wallets
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If the equation does not balance, the transaction is cancelled. This check is done twice: once by the application and once by database triggers, so even a bug or a manual database edit cannot silently create money.

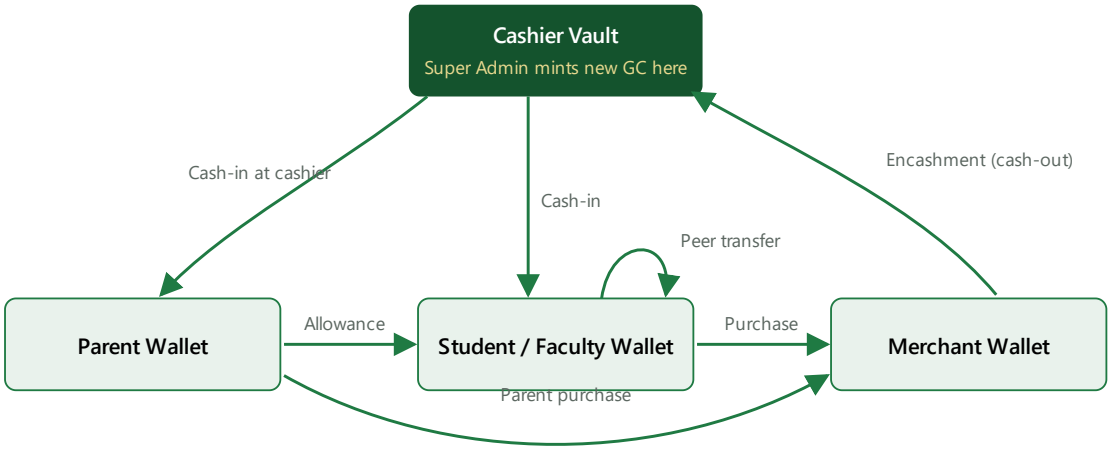


Figure 2. How GenCoins move between the vault and wallets

5.2 Core Money Flows

Flow	Who starts it	What happens
Minting	Super Admin	Raises the circulation cap and adds the new GC to the vault. Requires a reason and the mint PIN, and is limited by a monthly soft limit (50,000) and hard limit (500,000).
Cash-in (top-up)	Finance cashier, or a merchant	A user hands cash to the cashier (or a stall). GC equal to the amount is moved from the vault to the user's wallet. A 1% system fee is charged; merchant-assisted loading adds a 2% merchant cut.
Purchase	Student, faculty or parent	GC moves from the buyer's wallet to the merchant's wallet when a QR order or cart is paid. Stock is reduced automatically.
Peer transfer	Student / faculty	GC moves from one student wallet to another, found by school ID, within daily limits.
Allowance	Parent	GC moves from the parent's wallet to a linked child's wallet instantly, with no extra fee.
Refund / return	Merchant	Reverses a completed student payment, returning GC from the merchant to the student.
Encashment (cash-out)	Merchant requests, Finance releases	The merchant asks to convert earnings to cash. When finance releases the cash, the GC returns from the merchant wallet to the vault.
Fee-waiver claim	Student or parent	A school-approved credit is claimed once and added to the student's wallet (the cap is increased for it).

5.3 Typical Purchase at the Canteen

1. The student adds items to a cart by scanning a product QR code or typing its item code. A cart can only contain items from one stall.
2. The cart is submitted as a pending order; no money moves yet.
3. At the counter the merchant confirms the order, or the student scans the stall's POS QR code.
4. The system re-checks stock, the student's balance, wallet freeze status, daily spending limit and enrollment status.
5. The CirculationEngine moves the GC from the student's wallet to the merchant's wallet, reduces stock, and records the transaction and any maintenance fee.
6. Both sides get a notification and a receipt, and the parent sees the purchase in their activity feed.

5.4 Canteen Stall Application Lifecycle

1. **Apply.** A prospective merchant views the public stall map, picks a preferred stall and submits personal and business details with five documents (photo, business permit, sanitary permit, clearance, GJC requirements).
2. **Verification meeting.** The system automatically books a meeting in the next free weekday slot (09:00 to 16:00, skipping holidays) and emails the applicant.
3. **Review.** Finance reviews the application, may reschedule, reject (archived and reactivatable later) or continue.
4. **Contract and payment.** Finance uploads the signed contract and records the two-month deposit and one-month advance rent.
5. **Award.** The stall is awarded and the system creates the owner account, merchant wallet, lease and stall assignment, then emails the login credentials.

5.5 School Year and Enrollment

Wallet access for students depends on their enrollment status for the active school year. Enrolled students have full access; not-enrolled students are read-only; dropped, transferred or graduated students are blocked. At year rollover the system snapshots all balances and carries enrollment forward; no money is created or removed. Faculty are never affected by enrollment.

6. Features by Role

6.1 Student and Faculty Portal

Dashboard and Wallet

- Current balance in GC and pesos
- Recent transactions and account status
- Notifications with unread counter

Shopping and Payment

- Cart built by QR scan or item code
- Pay a merchant's POS QR order
- Checkout with stock, balance and limit checks
- Digital receipts for every purchase

Transfers

- Send GC to another student by school ID
- Recipient preview before sending
- Minimum amount and daily limit enforced

History and Receipts

- Full transaction history, filterable by school year
- Detailed transaction view with secure links

Fees and Waivers

- View maintenance fees charged
- Claim a posted fee-waiver credit once (students only)

Profile and Security

- Edit personal details and profile photo
- Change password, link guardian
- Faculty see a "Faculty Portal" label and log in by Employee ID

6.2 Parent Portal

Child Linking

- Link a child by school ID
- Supervise several children from one account
- Unlink a child

Allowance

- Send GC from the parent wallet to a child instantly
- No extra allowance fee

Spending Controls

- Freeze or unfreeze a child's wallet
- Set a daily spending limit
- Set a low-balance alert threshold

Activity Monitoring

- See each child's purchases, transfers and fees
- Filter activity and open transaction details

Alerts

- Low-balance alerts for linked children
- Mark alerts as read

Own Wallet and Shopping

- Parents have a separate wallet
- Buy at canteen stalls by cart or POS QR
- Claim a fee waiver on a child's behalf

6.3 Merchant Portal

Owner features apply to the stall owner. Staff can use the POS, inventory and transaction pages for their owner's stall, but cannot manage staff, money or the contract.

Point of Sale (Owner + Staff)

- Build an order and generate a 5-minute QR code with short code
- Look up a student and charge their wallet directly
- Pending order queue, order details and voiding

Inventory (Owner + Staff)

- Products with SKU, category, price, stock and low-stock alert
- Availability toggle and printable menu with QR codes
- Automatic check against restricted products

Sales and Transactions

- Sales history, voided transactions, receipts
- Returns and refunds to students
- Remittance reports by period

Wallet Loading

- Load GC to students or parents for cash received at the stall
- Merchant earns a 2% cut

Encashment (Owner)

- Request conversion of earnings to cash
- Track request status until released

Staff Management (Owner)

- Create, edit, deactivate staff
- Reset staff passwords, soft-remove staff

Stall and Lease (Owner)

- View contract and rent schedule, due dates and balance
- Rent reminders before due date

Business Profile (Owner)

- Stall name, logo, contact details
- Upload compliance documents
- Export a data backup

6.4 Finance / Administrator Portal

Dashboard

- Totals for wallets, vault and circulation
- Activity charts and pending queues
- Live circulation health widget

Cash Management

- Cash-in to student or parent wallets
- Release merchant encashments
- Cash-in history and daily cash records

User Management

- Create and view students, faculty, parents and merchants
- Suspend (up to 365 days), ban or lift restrictions
- Reset student passwords

Maintenance and Import

- Signed CSV templates for bulk student import
- Preview before import, automatic credential emails
- Direct merchant provisioning

Stall Applications

- Review queue, meeting schedule, rejections
- Contract upload, deposit and advance recording
- Award stall and auto-create merchant

Merchants and Leases

- Merchant directory with stall, lease and rent status
- Record or void rent payments
- Merchant details and compliance documents

Fees and Waivers

- Maintenance fee collection with receipts
- Fee-waiver credits: assign, post, print
- Bulk waiver import by CSV

School Years

- Create and activate school years
- Upload enrollment rosters and apply statuses
- Year rollover and graduation handling

Transactions and Reports

- Searchable transaction ledger
- Export filtered transactions and fees

Settings

- Transfer and top-up limits
- Maintenance rates, waiver limits
- Restricted-product thresholds

Super Admin only

- **Economy / Minting:** increase the circulation cap with PIN, reason and limits; set or change the mint PIN.
- **Audit Log:** browse, filter and verify the integrity of the hash-chained audit trail.
- **Backups:** generate full system backups.
- **Privileged account actions:** selected destructive or status-changing operations such as un-marking graduation.

7. Supporting Features

These features are not tied to a single page but work across the whole system.

Feature	What it does
Circulation Engine	The single gateway for moving money. It locks the books, moves GC, writes a ledger entry and verifies the closed-loop equation before saving.
Minting Guard	Protects money creation with Super Admin checks, a PIN and monthly soft/hard limits, and logs every attempt.
Circulation Widget	Dashboard panel that shows the cap, vault and wallet totals and whether the books balance.
Audit Trail	Append-only log of logins, money movements and admin actions, stored through a separate database connection. Each entry contains a SHA-256 hash of the previous one, so any edit or deletion breaks the chain and is detected. Passwords and secrets are redacted.
Notifications	In-app notifications for payments, transfers, allowances, rent, applications and alerts, with unread counters.
Email Service	Sends credentials, meeting schedules and notices. Messages are queued and sent by a background worker with three retries; if queuing fails it sends directly.
Restricted Products	Finance lists banned items. Product names are normalized (letter look-alikes, spacing, symbols) and matched, so variations are still caught. Repeated violations can suspend a merchant.
Maintenance Fees	Small configurable fees accrue on qualifying payments and transfers, are collected by finance with receipts, and are voided if the payment is refunded.
Fee-Waiver Credits	School-approved credits that move through assignment, signed-document posting and a one-time claim by the student or a linked parent.
Suspensions and Bans	Timed suspensions expire automatically; bans last until lifted. A restriction freezes the wallet without overriding a separate parent freeze.
Stall Map	Public map of the canteen stalls (rows A and B) showing which are available.
Rent Accounting	Splits each lease into monthly periods, applies payments, calculates balance and next due date, and sends reminders seven days before due.
Secure Documents	Uploaded permits, contracts and waivers are stored outside public access and served only through a checked document viewer.
Printable Outputs	PDF/printable receipts, fee-waiver forms, maintenance fee receipts, menus with QR codes and backups.
Profile Photos	JPEG, PNG or WebP photos up to 2 MB for any user.
Remember Me	Optional 30-day login using a rotating, hashed token.

8. Security and Controls

Area	Control
Authentication	Passwords stored as secure hashes; session ID regenerated on login; forced password change on first login and after resets; inactive, locked, suspended or banned accounts cannot log in.
Authorization	Every page and endpoint checks the user's role. Merchant staff can only reach their owner's resources; parents only their linked children. Sensitive finance actions require the Super Admin sub-role.
Financial integrity	Database transactions and row locks on every money movement; double-checked closed-loop equation in PHP and database triggers; minting PIN and limits.
Request protection	CSRF tokens on payments, transfers, allowances, waiver claims and settings; prepared statements against SQL injection; signed links for transaction details.
Uploads	File type and size checks, random draft names, storage outside the public folder, path-traversal protection on the document viewer.
Web server	Configuration, database, include and upload folders are blocked from direct access; directory listing disabled; security headers set.
Accountability	Tamper-evident audit trail with redaction, request IDs, actor, target, outcome and hashed session reference.

9. Data Overview

The main groups of data stored by GenPay:

Group	Main tables
Identity	users, role, student_info, parents, parent_student_links
Wallets and money	student_wallets, parent_wallets, merchant_wallets, system_settings (cap and vault), transactions, p2p_transfers, cap_increase_log, fee_revenue_log
Commerce	merchant_inventory, cart_orders, merchant_qr_orders, sales_remittances, encashment_requests, restricted products
Stalls and tenancy	stall_applications, archived_rejections, merchant_leases, merchant_rent_payments, merchant_documents
School and fees	school_years, student_enrollments, enrollment_roster_staging, maintenance_fees, maintenance_fee_collections, fee_waiver_credits
System	app_settings, notifications, parent_alerts, systemic_audit_trail

> The **users** table is the hub: each user has one role and, depending on that role, a student, parent or merchant wallet. The **transactions** table is the single ledger for every money movement.

10. Business Rules and Default Limits

Rule	Value
Exchange rate	1 GenCoin = ₱10.00 (fixed)
System service fee on cash-in	1% of the amount
Merchant cut on merchant-assisted loading	2% of the amount
Allowance fee (parent to child)	None
Minting soft limit / hard limit (per month)	50,000 / 500,000; PIN always required
Peer transfer minimum / daily limit (default)	1 / 5,000 (configurable)
POS QR order validity	5 minutes
Cart rule	Items from one merchant only
Maximum timed suspension	365 days
Stall award requirement	2 months deposit + 1 month advance
Verification meeting slots	09:00, 10:00, 11:00, 13:00, 14:00, 15:00, 16:00 on weekdays
Rent reminder window	7 days before due date (default)
Application draft cleanup	After 48 hours
Upload size limits	Documents 5 MB, profile photos 2 MB
Remembered login	30 days
Enrollment statuses	Enrolled, Not enrolled, On leave, Dropped, Transferred, Graduated

11. System Requirements and Deployment

Server

- Apache with mod_rewrite
- PHP 8.1+ with PDO MySQL, mbstring, fileinfo, openssl
- MySQL 8 / MariaDB 10.5+
- SMTP account for email (e.g. Gmail on port 587)
- Timezone: Asia/Manila

Client

- Any modern browser (Chrome, Edge, Firefox, Safari)
- Works on phones, tablets and desktops
- Camera access for QR scanning
- Internet access for hosted UI libraries

Configuration

Database, SMTP, audit-account and signing secrets are supplied through environment variables or local configuration files (`local_config.php`, `audit_config.php`, `mail_config.php`) that are kept out of version control. Database tables are created and upgraded automatically on first use.

12. Limitations and Future Enhancements

Current Limitations

- No online payment gateway or bank integration; all cash enters and leaves through the school cashier or a stall.
- No scheduled background jobs; expiries, reminders and clean-ups run when related pages are opened.
- No recurring (automatic) allowance; parents send allowances manually.
- No public self-registration or self-service password recovery; accounts and resets are handled by finance.
- Enrollment is uploaded by CSV; there is no live link to a registrar system.
- Web-only; no native mobile app or offline mode.

Possible Enhancements

- Online top-up through e-wallets such as GCash or Maya.
- Scheduled recurring allowances and automated reminders through a cron scheduler.
- Self-service password reset by email OTP.
- Registrar integration for real-time enrollment status.
- Installable mobile app (PWA) with push notifications.
- Nutrition and spending analytics for parents.

13. Glossary

Term	Meaning
GenCoin (GC)	The digital currency of GenPay. 1 GC = ₱10.
Circulation cap	Total GenCoin allowed to exist in the system.
Cashier vault	GenCoin held by the school cashier and not yet in any wallet.
Minting	Raising the circulation cap to create new GenCoin (Super Admin only).
Cash-in / Top-up	Exchanging physical cash for GenCoin in a wallet.
Encashment	A merchant exchanging GenCoin earnings back to physical cash.
Allowance	GenCoin sent from a parent to a linked child.
Remittance	A merchant's sales report boundary for a period; it does not move money.
Maintenance fee	A small accrued charge on qualifying transactions, collected by finance.
Fee-waiver credit	A school-approved credit claimed once into a student's wallet.
Closed loop	The rule that GenCoin only moves between the vault and wallets and is never created or lost.
Audit trail	The tamper-evident, append-only record of system activity.
POS	Point of Sale, the merchant's checkout screen.